Semester 3													
Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits	Examination Scheme					
							Theory			Practical		Total
							MS	IA	ES	LW	LE/Viva	
24INT151T	Civic and Social Service Internship	0	0	2	2	1	--	100	--	--	--	100
24MA201T	Mathematics - III	3	1	0	4	4	25	25	50	--	--	100
24CH202T	Introduction to Artificial Intelligence	3	0	0	3	3	25	25	50	--	--	100
24CH203T	Chemical Process Calculations	3	1	0	4	4	25	25	50	--	--	100
24CH204T	Mechanical Unit Operations	3	0	0	3	3	25	25	50	--	--	100
24CH204P	Mechanical Unit Operations Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH205T	Fluid Mechanics	3	0	0	3	3	25	25	50	--	--	100
24CH205P	Fluid Mechanics Laboratory	0	0	2	2	1	--	--	--	50	50	100
	Total	**15**	**2**	**6**	**23**	**20**						

Semester 4													
Course Code	Course Name	Theory	Tutorial	Practical	Hrs	Credits	Examination Scheme					
							Theory			Practical		Total
							MS	IA	ES	LW	LE/Viva	
24INT251T	Industrial Orientation	0	0	0	0	0	--	--	--	--	--	Audit
OE-1	Open Elective 1 (Inter-School)	3	0	0	3	3	25	25	50	--	--	100
24CH206T	Chemical Engineering Industry 4.0	2	0	0	2	2	25	25	50	--	--	100
24CH206P	Chemical Engineering Industry 4.0 Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH207T	Heat Transfer	3	0	0	3	3	25	25	50	--	--	100
24CH207P	Heat Transfer Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH208T	Chemical Process Technology	3	0	0	3	3	25	25	50	--	--	100
24CH208P	Chemical Process Technology Laboratory	0	0	2	2	1	--	--	--	50	50	100
24CH209T	Introduction to Numerical Methods for Chemical Engineers	3	1	0	4	4	25	25	50	--	--	100
24CH210T	Chemical Engineering Thermodynamics	3	0	0	3	3	25	25	50	--	--	100
24CH210P	Chemical Engineering Thermodynamics Laboratory	0	0	2	2	1	--	--	--	50	50	100
	Total	**17**	**1**	**8**	**26**	**22**						

24INT151T					Civic and Social Service Internship					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	100 (CE)	--	--	100

COURSE OBJECTIVES

- To sensitize students to the needs, problems and aspirations of the community.
- To engage students in structured civic and social service activity outside the classroom.
- To develop empathy, teamwork, communication and leadership through field engagement.
- To document and reflect upon the social impact of the service rendered.

COURSE CONTENT / ACTIVITIES

1. Identification of a host organization / community group and formulation of the service plan in consultation with the faculty mentor.
2. Field engagement in civic, social, environmental or educational service for the prescribed duration.
3. Maintenance of a daily/weekly logbook of activities, observations and reflections.
4. Submission of an internship report describing the organization, activities performed, outcomes and learning.
5. Presentation and viva-voce before the departmental evaluation committee.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the needs and problems of the community in a structured manner.
CO2 - Understand individual responsibility towards society and civic life.
CO3 - Apply disciplinary and interpersonal skills in a real community setting.
CO4 - Analyse the social, environmental and economic dimensions of the issue addressed.
CO5 - Evaluate the impact of the service rendered and propose sustainable improvements.
CO6 - Communicate the work carried out through an effective report and presentation.

TEXT / REFERENCE BOOKS

1. Pandit Deendayal Energy University. (2024). *Internship guidelines and logbook*. PDEU.
2. Government of India. (2023). *National Service Scheme manual*. Ministry of Youth Affairs and Sports.

EVALUATION

Evaluation is by Continuous Evaluation (CE) for 100 marks, comprising the logbook, internship report, presentation and viva-voce assessed by the departmental committee.

24MA201T					Mathematics - III					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

- To apply Fourier analysis for solving applications in chemical and allied engineering branches.
- To familiarize students with a variety of engineering problems that can be analyzed using the properties of Fourier transform techniques.
- To provide a broad coverage of mathematical techniques that are widely used to obtain analytical solutions to partial differential equations of first and second order.
- To introduce various applications of partial differential equations in many fields of science and engineering.

Unit I: FOURIER SERIES **Hours: 13**

Periodic functions; odd and even functions; Euler's formulae for Fourier series in an interval of 2π; change of interval; Dirichlet's conditions; half-range sine and cosine series; complex Fourier series; Parseval's identity and its applications.

Unit II: FOURIER TRANSFORM **Hours: 13**

Fourier integral theorem; sine and cosine integrals; Fourier transform; Fourier sine and cosine transforms; properties; inverse Fourier transform; applications of the Fourier transform to problems on one-dimensional and two-dimensional heat flow.

Unit III: PARTIAL DIFFERENTIAL EQUATIONS OF FIRST ORDER **Hours: 13**

Formation of partial differential equations (PDEs); solutions of PDEs of first order; Cauchy problem for first order PDEs; Lagrange's method; Charpit and Jacobi methods for solving first order nonlinear PDEs.

Unit IV: PARTIAL DIFFERENTIAL EQUATIONS OF SECOND ORDER WITH APPLICATIONS **Hours: 13**

Classification of second order PDEs; method of separation of variables; Fourier series solutions of the one-dimensional wave equation; one-dimensional heat equation; steady state solution of the two-dimensional heat equation; applications of PDEs to string and rod problems; finite and infinite plate problems and reaction engineering.

Max 52 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify real phenomena as models of partial differential equations.
CO2 - Develop the skills for solving boundary value problems arising in chemical engineering.
CO3 - Apply various analytical methods to obtain solutions to PDEs which occur in science and engineering.
CO4 - Formulate and solve physical problems involving partial derivatives.
CO5 - Apply the techniques learnt to analyse a comprehensive model related to chemical engineering.
CO6 - Demonstrate the ability to use mathematical arguments to describe real-world problems in science and engineering.

TEXT / REFERENCE BOOKS

1. Rao, K. S. (2010). *Introduction to partial differential equations* (2nd ed.). PHI Learning.
2. Amaranath, T. (2003). *An elementary course in partial differential equations* (2nd ed.). Narosa Publishing House.
3. Evans, L. C. (1998). *Partial differential equations* (Graduate Studies in Mathematics, Vol. 19). American Mathematical Society.
4. Grewal, B. S. (2017). *Higher engineering mathematics* (44th ed.). Khanna Publishers.
5. Kreyszig, E. (2016). *Advanced engineering mathematics* (10th ed.). John Wiley & Sons.
6. Jain, R. K., & Iyengar, S. R. K. (2002). *Advanced engineering mathematics* (3rd ed.). Narosa Publishing House.

24CH202T					Introduction to Artificial Intelligence					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To provide the link between artificial intelligence and real-time processes in various practical applications.
- To provide the basic concepts of machine learning with examples linked to automation-based industrial procedures.
- To pitch the course at a level such that all students aspiring to be a part of artificial intelligence, directly or indirectly, acquire these concepts.
- To apply the concepts of artificial intelligence to the design, stability analysis and controllability of chemical processes.

Unit I: FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE **Hours: 7**

Introduction to AI: definitions, history, importance, ethics, challenges and future directions; evolution of AI technologies and their impact across various sectors; relevance to chemical engineering. Agents of artificial intelligence: introduction to AI agents, environments and types of agents (from simple reflex agents to learning agents), focusing on their potential applications in process control and optimization; understanding the working principle of fuzzy systems and artificial neural networks.

Unit II: ADVANCED AI TECHNOLOGIES AND METHODS **Hours: 12**

Evolutionary computing (EC): principles of evolutionary computing with a focus on genetic algorithms, covering their structure and application to optimization problems found in chemical engineering. Natural language processing (NLP): introduction to NLP and its applications in analyzing scientific literature, patents and documentation within the chemical engineering domain.

Unit III: OPTIMIZATION TECHNIQUES IN AI **Hours: 11**

Genetic algorithms (GA): detailed exploration of GAs for solving complex optimization problems in chemical engineering. Simulated annealing (SA) and particle swarm optimization (PSO): introduction and application of SA and PSO in finding optimal solutions under constraints typical of chemical processes; supervised learning problems (SLP): classification and regression methods including k-nearest neighbour (KNN) and decision trees for prediction and decision making; unsupervised learning problems (USLP): clustering techniques including k-means clustering for pattern discovery in chemical engineering datasets.

Unit IV: APPLICATION OF AI IN CHEMICAL ENGINEERING **Hours: 9**

AI in process control: utilizing AI for improving process control strategies, including predictive maintenance and real-time optimization. AI in material science: leveraging AI for material discovery and development, emphasizing prediction of material properties and formulations. AI in environmental engineering: application of AI in waste treatment, emission control and environmental monitoring; sensor networks, IoT and data acquisition for environmental monitoring; regression models, artificial neural networks (ANN) and random forest for predicting treatment performance and effluent quality.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand the basics, history and evolution of AI, differentiating it from traditional computing.
CO2 - Identify and explain AI agents and their roles in chemical engineering.
CO3 - Apply fuzzy systems and neural networks in chemical engineering problem solving.
CO4 - Design and implement genetic algorithms and deep learning for chemical engineering optimization.
CO5 - Utilize AI optimization techniques to enhance chemical engineering processes.
CO6 - Critically assess the impact of AI and its ethical considerations in chemical engineering.

TEXT / REFERENCE BOOKS

1. Winston, P. H. (2004). *Artificial intelligence* (3rd ed.). Addison-Wesley Publishing Company.
2. Nilsson, N. J. (2014). *Principles of artificial intelligence*. Springer Heidelberg.
3. Russell, S., & Norvig, P. (2009). *Artificial intelligence: A modern approach* (3rd ed.). PHI Learning.
4. Nilsson, N. J. (2010). *The quest for artificial intelligence*. Cambridge University Press.

24CH203T					Chemical Process Calculations					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

- To learn to use the correct unit conversions and dimensions and use them in process calculations.
- To synchronize the understanding of stoichiometry with different unit operations and processes.
- To handle material balances for systems with and without reactions and with recycle/bypass operations.
- To accustom students to different forms of energy and terminologies of heat and energy and to perform energy balances.

Unit I: BASIC CONCEPTS OF CHEMICAL PROCESSES **Hours: 13**

Units and dimensions; basic chemical calculations; dimensional consistency; unit processes and unit operations; concepts of atomic weight, equivalent weight and mole; composition of solids, liquids and solutions (weight percent, mole percent, molarity, normality); other expressions for concentration; average molecular weight and density; gaseous mixtures; ideal gas laws and their applications; Raoult's law; Henry's law; Amagat's law and Dalton's law; humidity and saturation; unit conversion from mg/m³ to ppm for TLV; conversion of flammability limits from percentage to ppm.

Unit II: MATERIAL BALANCES **Hours: 14**

Material balance for non-reacting multi-phase systems; solving material balance problems without chemical reactions for unit operations such as absorption and stripping, distillation, extraction and leaching, drying, evaporation, crystallization and mixing/blending; material balance for systems with recycle and bypass; material balances on systems with reactions; concept of limiting and excess reactants, percentage conversion, selectivity and yield; properties of gases, liquids and solids; ideal and non-ideal gas calculations; equations of state; phase equilibria for ideal and real mixtures; humidity and saturation.

Unit III: ENERGY BALANCES **Hours: 14**

Terminologies; specific heat capacity; enthalpy; heat of formation, combustion, reaction and solution; energy balances without reactions and for chemical reactions; isothermal systems; adiabatic systems; simultaneous material and energy balances; unsteady state material and energy balances; exothermic reactions; adiabatic temperature rise and thermal runaway; latent heat of vaporization and BLEVE; case study of energy balance: MIC and non-MIC route.

Unit IV: FUEL AND COMBUSTION **Hours: 11**

Types of fuels; proximate and ultimate analysis of fuel; combustion theory; heating value of fuels; calculations involving theoretical and excess air; heat and material balances of combustion processes; analysis of products of combustion; calorific value - HCV and LCV; limiting oxygen concentration.

Max 52 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Recall the correct unit conversions and dimensions for different variables for chemical calculations.
CO2 - Relate stoichiometry to material and energy balances for different unit operations and processes.
CO3 - Solve material balances of process systems with and without reactions, including recycle/bypass operations.
CO4 - Describe energy changes in liquids and gases accompanying various chemical reactions with the terms used to associate energy changes in different phases.
CO5 - Evaluate fuel quality and the requirements of gases in combustion.
CO6 - Elaborate and build the mass and energy requirements of a process.

TEXT / REFERENCE BOOKS

1. Himmelblau, D. M., & Riggs, J. B. (2012). *Basic principles and calculations in chemical engineering* (8th ed.). PHI Learning.
2. Hougen, O. A., Watson, K. M., & Ragatz, R. A. (2004). *Chemical process principles: Part 1* (2nd ed.). John Wiley & Sons.
3. Narayanan, K. V., & Lakshmikutty, B. (2006). *Stoichiometry and process calculations*. Prentice Hall of India.
4. Bhatt, B. I., & Thakore, S. B. (2010). *Stoichiometry* (5th ed.). Tata McGraw Hill.

24CH204T					Mechanical Unit Operations					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To understand the importance and applications of mechanical unit operations in process industries.
- To learn the techniques of solid characterization, size reduction and handling operations.
- To understand solid-fluid operations including filtration, mixing, agitation and transportation.
- To compare solid transportation methods and equipment for appropriate application.

Unit I: SOLID CHARACTERIZATION - CLASSIFICATION AND SIZE REDUCTION **Hours: 12**

Introduction: unit operations and their role in chemical industries. Properties and handling of particulate solids: characterization of solid particles, properties of masses of particles, classification of solids, sieving and screening equipment, mixing of solids. Size reduction: laws of size reduction, energy relationships in size reduction, power requirement, work index; methods of size reduction; classification of equipment - crushers, grinders, disintegrators for coarse, intermediate and fine grinding, ultrafine grinders.

Unit II: SOLID-FLUID OPERATIONS **Hours: 9**

Gravity settling: free and hindered settling, sedimentation, thickening, elutriation, double cone classifier, rake classifier, bowl classifier. Centrifugal separation: continuous centrifuges, design of basket centrifuges. Flow through packed beds: fixed bed and fluidized bed; pneumatic conveying; industrial dust removing equipment; cyclones and hydrocyclones; electrostatic and magnetic separators; heavy media separation; flotation.

Unit III: SOLID-FLUID SEPARATION - FILTRATION **Hours: 9**

Theory of filtration; batch and continuous filters; flow through filter cake and filter media; compressible and incompressible filter cakes; filtration equipment - selection, operation and design of filters; optimum cycle of operation; filter aids; safety issues related to high-speed rotating equipment.

Unit IV: STORAGE AND TRANSPORTATION OF SOLIDS **Hours: 9**

Agitation and mixing of liquids; storage and conveying of solids - bunkers, silos, bins and hoppers; transportation of solids in bulk: different types of conveyors and their performance characteristics.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Choose the techniques of mechanical operations to meet the needs of chemical industries.
CO2 - Select suitable size reduction equipment and solid-solid separation methods.
CO3 - Understand fluid flow through packed and fluidized beds.
CO4 - Identify different types of mixing, agitation and conveying of solids and estimate the power requirement.
CO5 - Determine the static and dynamic principles of separation for ores in chemical industries.
CO6 - Design filtration equipment by considering constant and variable pressure governing equations.

TEXT / REFERENCE BOOKS

1. Coulson, J. M., & Richardson, J. F. (2006). *Particle technology and separation processes* (5th ed.). Elsevier.
2. McCabe, W. L., Smith, J. C., & Harriott, P. (2014). *Unit operations of chemical engineering* (7th ed.). McGraw Hill Education.
3. Badger, W. L., & Banchero, J. T. (2001). *Introduction to chemical engineering*. McGraw-Hill Education.
4. Foust, A. S., Wenzel, L. A., Clump, C. W., Maus, L., & Andersen, L. B. (2008). *Principles of unit operations* (2nd ed.). Wiley.
5. Swain, A. (2017). *Mechanical operations*. McGraw Hill Education.

24CH204P					Mechanical Unit Operations Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To analyse the complex engineering problem behind the working of mechanical operation equipment.
- To investigate data from experiments and apply the knowledge of data interpretation, design of experiments and statistical analysis to arrive at a solution by comparison with industrial equipment.
- To understand proper engineering practices in a chemical process industry with emphasis on health, safety, society, environment and sustainable development.
- To develop the ability to work in a team and report experimental findings effectively.

LIST OF EXPERIMENTS

1. Screen analysis and determination of different mean diameters using a sieve shaker.
2. Size reduction and characteristics of a jaw crusher.
3. Size reduction and characteristics of a roll crusher.
4. Size reduction and characteristics of a ball mill.
5. Size reduction and characteristics of a hammer mill.
6. Determination of the effectiveness of a screen.
7. Study of the fluidization process and calculation of minimum fluidization velocity.
8. To determine the collection efficiency of a cyclone separator.
9. To determine the average specific cake resistance, filter medium resistance and cake weight for a horizontal filter press.
10. To study the operational characteristics of a top-driven centrifuge.
11. To plot the mixing index versus mixing time curve for a ribbon blender.
12. To study the working of a screw conveyor.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Relate the classical knowledge of basic sciences and mathematical correlations to solve equations based on mechanical unit operations in the laboratory as part of a team.

CO2 - Demonstrate the design and working of mechanical unit operations and appreciate the importance of health, hygiene, safety and sustainable development.

CO3 - Apply data interpretation techniques, plot graphs based on experiments performed and use design of experiments for data optimization.

CO4 - Take part in the successful completion of an experiment as a team building exercise and follow ethical practices judiciously.

CO5 - Learn the importance of effective written and oral communication in preparing laboratory observations and reports.

CO6 - Elaborate the importance of different mechanical unit operations as a lifelong learning exercise in order to comprehend technological change.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Mechanical unit operations laboratory manual*. Pandit Deendayal Energy University.
2. Swain, A. (2017). *Mechanical operations*. McGraw Hill Education.
3. McCabe, W. L., Smith, J. C., & Harriott, P. (2014). *Unit operations of chemical engineering* (7th ed.). McGraw Hill Education.
4. Badger, W. L., & Banchero, J. T. (2001). *Introduction to chemical engineering*. McGraw-Hill Education.
5. Foust, A. S., Wenzel, L. A., Clump, C. W., Maus, L., & Andersen, L. B. (2008). *Principles of unit operations* (2nd ed.). Wiley.

24CH205T					Fluid Mechanics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

- To develop basic concepts of fluid flow and learn flow analysis leading to problem-solving skills.
- To understand the concept of a flow system involving various flow types.
- To carry out design and analysis of fluid transportation devices and systems, including agitation and mixing.
- To understand the principles of flow measuring systems for various flow situations.

Unit I: FLUID STATICS AND ITS APPLICATION **Hours: 9**

Introduction to fluid mechanics: definition and properties of fluids; fluid statics - pressure distribution in a fluid; dimensional analysis - different methods of dimensional analysis applied to fluid flow problems; types of flow; relationship between shear stress and pressure gradient; fluid flow in different channels; Hagen-Poiseuille equation; losses in pipes and fittings; pipe joints, fittings, bends and erosion issues.

Unit II: BASIC EQUATIONS OF FLUID FLOW **Hours: 12**

Navier-Stokes equations; continuity equation; Bernoulli equation; Euler equation; equation of motion; Darcy-Weisbach equation for frictional head loss; friction factor; Moody diagram; velocity profile and boundary layer calculations for turbulent flow; detailed leak flow rate calculations (integrating the time derivative, for both liquids and gases).

Unit III: TRANSPORTATION OF FLUIDS **Hours: 9**

Handling of fluids: pumps, compressors and blowers for handling different fluids; standards for pumps, compressors and blowers; valves, pipe fittings and their standards; power requirement for flow; piping layout and economical pipe diameter; priming of pumps; safety issues in pumping volatile substances.

Unit IV: METERING OF FLUIDS **Hours: 9**

Flow metering devices: orifice meter, venturi meter, rotameter, pitot tube, anemometer; flow through open channels such as notches, weirs and nozzles; vacuum producing devices; two-phase flow - basic principles and applications.

Max 39 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Define the principles of fluid mechanics operations.
CO2 - Explain the theoretical importance and relevance of fluid flow in the chemical process industry.
CO3 - Identify and apply the theoretical concepts of fluid flow in the chemical process industry.
CO4 - Comprehend and analyse fluid mechanics problems with the application of conservation principles of mass, energy and momentum.
CO5 - Evaluate fluid mechanics problems with the application of conservation principles of mass, energy and momentum.
CO6 - Design fluid transportation systems such as pumps, compressors and pipe networks and choose fluid transportation devices for process applications.

TEXT / REFERENCE BOOKS

1. McCabe, W. L., Smith, J. C., & Harriott, P. (2014). *Unit operations of chemical engineering* (7th ed.). McGraw Hill.
2. Foust, A. S., Wenzel, L. A., Clump, C. W., & Andersen, L. B. (2008). *Principles of unit operations* (2nd ed.). Wiley.
3. Modi, P. N., & Seth, S. M. (2009). *Hydraulics and fluid mechanics* (19th ed.). Standard Book House.
4. White, F. M. (2008). *Fluid mechanics* (6th ed.). Tata McGraw-Hill.
5. Pritchard, P. J., & Mitchell, J. W. (2016). *Fox and McDonald's introduction to fluid mechanics* (9th ed.). John Wiley & Sons.

24CH205P					Fluid Mechanics Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hours/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

- To learn to perform experiments and analyse and interpret the observations yielded.
- To compare the results of analytical models introduced in lectures with the actual behaviour of real fluid flows.
- To discuss and practise standard measurement techniques of fluid mechanics and their applications.
- To learn to design experimental models, combining observations and theoretical understanding.

LIST OF EXPERIMENTS

1. Study of flow regimes using Reynolds' apparatus.
2. Study of Bernoulli's equation.
3. Determination of viscosity by efflux time measurement.
4. Study of friction factor in closed conduits.
5. Study of minor losses and determination of equivalent length of pipe fittings.
6. Study of a venturi meter.
7. Study of an orifice meter.
8. Calibration of a rotameter.
9. Study of a pitot tube.
10. Characteristics of a centrifugal pump.
11. Study of friction factor in annular flow.
12. Determination of viscosity by Stokes' law.

Max 26 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Relate the theoretical and practical concepts of fluid mechanics used in industry.

CO2 - Compare the results of analytical study introduced in lectures with the actual behaviour of real fluid flows and draw correct and sustainable conclusions.

CO3 - Develop the ability to work in groups on small design projects appropriate to the course.

CO4 - Categorize ethical issues associated with decision making and professional conduct.

CO5 - Assess the ability to write clear laboratory reports.

CO6 - Take part in the successful completion of an experiment as a team building exercise and follow ethical practices judiciously.

TEXT / REFERENCE BOOKS

1. Department of Chemical Engineering. (2024). *Fluid mechanics laboratory manual*. Pandit Deendayal Energy University.
2. McCabe, W. L., Smith, J. C., & Harriott, P. (2014). *Unit operations of chemical engineering* (7th ed.). McGraw Hill.
3. White, F. M. (2008). *Fluid mechanics* (6th ed.). Tata McGraw-Hill.
4. Pritchard, P. J., & Mitchell, J. W. (2016). *Fox and McDonald's introduction to fluid mechanics* (9th ed.). John Wiley & Sons.